INF2009F -- 2025

Problems in Software Development & Software Development Methodologies

Exercise 1 ---- 15 Marks

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Marks	/15

IMPORTANT NOTE:

- **Copied or Plagiarised answers will get 0 and student will be referred to Maureen Tanner and Ayanda Pekane for further actions**
- **Please upload your completed exercise on Amathuba by the 24th February 2024 at 9:00am**

1. Explain <u>in your own words</u>, why IT Projects fail (3 marks)

a. IT Projects fail because of the handling of projects in an unstructured and informal manner, usually, project requirements keep on changing and poor project requirements definition leads to changes that are not evaluated and anticipated which leads to projects going over the budget or losing money and eventually leads to the project's failure.

b. IT projects also fail because of a project's lack of flexibility and ability to adapt to changing "market" trends which then leads to undetected inconsistences and failure to manage risks.

c. IT projects also fail because of the use of architecture that is not flexible, difficult to work with and that is prone to breaking or crashing when changes are made. This makes it hard for developers to make changes to the project and leads to the project crashing unexpectedly and to the developers spending more time trying to fix bugs which is costly and time-consuming and, in the end, they may end up with a project that does not meet the requirements, crashes randomly and thus leading to it failing.

2. What are the disadvantages of using a plan-based approach (Traditional SDLC) to software development? (Describe each disadvantage in your own words) (3 marks)

1. A plan-based approach is not innovative enough and has handovers that are time-consuming to handle complex software development projects which makes it hard for programmers to update the requirements to meet the current needs of the customers and due to this lack of adaptability it may fail when it is implemented, thus making it unsuitable for projects where customer needs evolve over time.

2. In the plan-based approach, there is a lack of communication between stakeholders, and in the later stages when the stakeholders want to make changes it is difficult to make them which then affects the developer's scope and integrity of the project.

3. The plan-based approach only does testing at the end of each project which means that developers are working on something that they are not clear as to what the outcome will look like and can only detect any defects at the end of the project which wastes time and is expensive for the business to go back and fix, thus causing delays in the implementation of the system.

3. Give a summary of the Agile manifesto (3 marks)
(Hint – do some research on this)

The Agile Manifesto is a methodology created by a group of professional developers in 2001 with the intention of creating a software development process that is flexible and quick enough to anticipate and respond to changing requirements during project development. This is a process that develops software in an iterative and incremental way associated with 4 values prioritizing individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation and a response to change over following a plan. Moreover, the manifesto is guided by 12 principles emphasizing quick and early delivery of projects, customer satisfaction, flexibility, adaptability, teamwork and communication, simplicity, and continuous improvement that will allow developers to refine and modify the project with less effort and at a lower cost ensuring the delivery of a high-quality project.

4. **Use (in your opinion) 3 of the most significant of the 12 principles of Agile development to motivate why agile software development methodologies can resolve the problems introduced by plan-based (traditional SDLC) approaches? (6 marks)**

Development methodologies—Mike Chapple:
https://www.linkedin.com/learning/search?keywords=Systems%20development%20methodologies%20&u=70295562

1. **Customer Satisfaction through early and continuous delivery of valuable software:** Early and continuous delivery of valuable software is significant because it will create a state of trust and flexibility between the customer and the development team, the development team will get feedback on the correctness of their understanding of the product requirement through the customer meaningful feedback involvement since customers are the ones who will use the finished project, this will make the customers feel more confident when they 'touch' a basic fully or partially functioning version of a component in their product and will know what to expect while also help developers to make the necessary changes to fully meet the expectations of the customers which saves time and money.

2. **Build projects around motivated individuals:** Traditional SDLC often follows a top-down management approach, where developers are expected to execute detailed plans with little freedom of creative expression. Agile fosters self-organizing teams, empowering individuals to make decisions, take ownership of their work, and collaborate effectively. This leads to higher engagement, creativity, and productivity and will make the parties involved feel like an important part of the project and not just feel like the 'worker' and 'consumer', ultimately resulting in better software.

3. **Welcome changing requirements:** Plan-based development often focuses on strict documentation and upfront design, which can become outdated or impractical over time due to the ever-changing needs of customers. Agile prioritizes maintainable code, testing after a few stages, and best practices, ensuring that software remains relevant, adaptable, and high-quality. By continuously refining code and design, teams can respond to changes

more efficiently without spending money on going back after too many stages to fix and modify the old code.